

Email Spam Classification

1. Overview:

This report describes the steps and outcomes of using a Multinomial Naive Bayes algorithm to build an email spam classification model. Accurately identifying emails as "spam" or "ham" (not spam) based on their content is the goal.

2. Data Loading and Exploration:

The 'spam.csv' file contains the dataset that was used for this project. The following was discovered during the dataset's initial exploration:

Shape of the Dataset: The dataset contains 5572 rows and 2 columns.

Columns: 'Message' (email text content) and 'Category' (target variable).

Data Types: An object type is present in both columns.

Unique Values: The 'Message' column contains some duplicate entries, with 5157 unique messages and 2 unique categories ('ham' and 'spam').

Class Distribution: There are substantially more "ham" emails (4825) than "spam" emails (747), indicating an imbalance in the dataset.

3. Data Cleaning and Preprocessing:

The 'Message' column underwent the following preprocessing procedures to get the text data ready for model training:

Lowercase Conversion: To maintain uniformity, all text was changed to lowercase.

Removal of Punctuation: For classification purposes, punctuation marks were eliminated because they typically don't add to the text's meaning.

Stopword Removal: The NLTK library was used to eliminate common English stopwords, such as "the," "a," and "is." This makes it easier to cut through the clutter and concentrate on words that are more instructive.

The preprocessed text was saved in a new column called "Cleaned_Message."

4. Vectorization:

The TF-IDF (Term Frequency-Inverse Document Frequency) vectorization technique was used to transform the text data into a numerical format that the machine learning model could use.

TfidfVectorizer was created as an instance. A matrix with each row representing an email and each column representing a distinct word—the values of which indicate the word's importance in that email in relation to the entire dataset—was produced by

fitting and transforming the vectorizer on the 'Cleaned_Message' column. After preprocessing, the resulting TF-IDF matrix has the shape (5572, 9455), which indicates that there were 5572 emails and 9455 unique words (features).

5. Model Training:

To assess the model's performance on unseen data, the dataset was divided into training and testing sets.

Features (X): The matrix of TF-IDF.

Target (y): The 'Category' column ('ham' or 'spam').

Split Ratio: 25% for testing ($\text{test_size}=0.25$) and 75% for training.

Random State: For reproducibility, 42 was employed.

The resulting sets have the following shapes:

X_{train} : (4179, 9455)

X_{test} : (1393, 9455)

y_{train} : (4179,)

y_{test} : (1393)

6. Model Building:

For this classification task, a Multinomial Naive Bayes model was selected, which works well for text classification using discrete features such as word counts or TF-IDF scores. MultinomialNB was created as an instance. The training data (X_{train} and y_{train}) were used to train the model.

7. Model Evaluation:

Several common classification metrics were used to assess the trained model on the testing data (X_{test} and y_{test}):

Accuracy: The percentage of emails that are correctly classified.

Precision: The percentage of emails that were truly spam out of all those that were predicted to be spam.

Recall: What percentage of all emails that were truly spam did the model correctly identify?

F1-Score: A balanced indicator of the model's performance that is calculated as the harmonic mean of precision and recall.

Confusion Matrix: A table that summarizes the quantity of false positives, false negatives, true positives, and true negatives.

The following are the evaluation's findings:

Precision: 0.9670

Accuracy: 1.0000

F1-Score: 0.8589

Recall: 0.7527

Matrix of Confusion: [[1207, 0] [46, 140]]